SoccerGMR Standard Test (1036/1036) — threshold extra

Standard-trained checkpoints | Standard Test (1036/1036) | 544 positive / 492 negative

Value above; (method - same-backbone GMR) below. G-mIoU: 0-1; others: percent / percentage points.

Backbone	Method	Acc τ=0.4	Rej-P τ=0.4	Rej-R τ=0.4	G-mIoU@5 τ=0.4	Acc τ=0.5	Rej-P τ=0.5	Rej-R τ=0.5	G-mIoU@5 τ=0.5	Acc τ=0.6	Rej-P τ=0.6	Rej-R τ=0.6	G-mIoU@5 τ=0.6
Moment-DETR	**GMR (baseline)**	**52.51**	**0.00**	**0.00**	**0.0134**	**68.34**	**60.00**	**100.00**	**0.4847**	**68.34**	**60.00**	**100.00**	**0.4847**
	GMR + GEQ Frozen	68.34 (+15.83)	60.00 (+60.00)	100.00 (+100.00)	0.4847 (+0.4713)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4847 (0.0000)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4847 (0.0000)
	GMR + GEQ Full	68.15 (+15.64)	59.90 (+59.90)	99.59 (+99.59)	0.4832 (+0.4698)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4852 (+0.0005)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4852 (+0.0005)
	GMR + Multi	65.35 (+12.84)	59.65 (+59.65)	83.54 (+83.54)	0.4070 (+0.3936)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4837 (-0.0010)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4837 (-0.0010)
	GMR + Multi + GEQ Frozen	66.70 (+14.19)	59.87 (+59.87)	90.65 (+90.65)	0.4396 (+0.4262)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4837 (-0.0010)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4837 (-0.0010)
	GMR + Multi + GEQ Full	64.86 (+12.35)	59.44 (+59.44)	81.91 (+81.91)	0.4002 (+0.3868)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4847 (0.0000)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4847 (0.0000)
EaTR	**GMR (baseline)**	**65.35**	**69.28**	**48.58**	**0.2635**	**66.80**	**64.57**	**66.67**	**0.3438**	**70.08**	**63.46**	**87.20**	**0.4358**
	GMR + GEQ Frozen	65.93 (+0.58)	64.21 (-5.07)	63.82 (+15.24)	0.3295 (+0.0660)	69.11 (+2.31)	62.46 (-2.11)	87.60 (+20.93)	0.4367 (+0.0929)	69.21 (-0.87)	61.10 (-2.36)	96.75 (+9.55)	0.4758 (+0.0400)
	GMR + GEQ Full	66.12 (+0.77)	64.36 (-4.92)	64.23 (+15.65)	0.3319 (+0.0684)	70.95 (+4.15)	63.62 (-0.95)	90.65 (+23.98)	0.4512 (+0.1074)	69.31 (-0.77)	60.98 (-2.48)	98.17 (+10.97)	0.4822 (+0.0464)
	GMR + Multi	65.35 (0.00)	71.52 (+2.24)	44.92 (-3.66)	0.2487 (-0.0148)	66.80 (0.00)	66.16 (+1.59)	61.59 (-5.08)	0.3236 (-0.0202)	68.05 (-2.03)	63.86 (+0.40)	75.41 (-11.79)	0.3850 (-0.0508)
	GMR + Multi + GEQ Frozen	69.11 (+3.76)	66.80 (-2.48)	69.51 (+20.93)	0.3574 (+0.0939)	68.44 (+1.64)	62.95 (-1.62)	81.50 (+14.83)	0.4104 (+0.0666)	68.53 (-1.55)	61.59 (-1.87)	89.63 (+2.43)	0.4454 (+0.0096)
	GMR + Multi + GEQ Full	68.63 (+3.28)	66.34 (-2.94)	68.90 (+20.32)	0.3563 (+0.0928)	68.53 (+1.73)	63.09 (-1.48)	81.30 (+14.63)	0.4109 (+0.0671)	68.92 (-1.16)	61.77 (-1.69)	90.65 (+3.45)	0.4509 (+0.0151)
QD-DETR	**GMR (baseline)**	**65.93**	**67.96**	**53.46**	**0.2848**	**69.59**	**62.48**	**90.04**	**0.4485**	**69.59**	**61.30**	**97.56**	**0.4798**
	GMR + GEQ Frozen	68.73 (+2.80)	64.19 (-3.77)	77.24 (+23.78)	0.3920 (+0.1072)	69.88 (+0.29)	62.33 (-0.15)	92.48 (+2.44)	0.4592 (+0.0107)	69.11 (-0.48)	61.00 (-0.30)	96.95 (-0.61)	0.4766 (-0.0032)
	GMR + GEQ Full	69.31 (+3.38)	64.08 (-3.88)	80.49 (+27.03)	0.4082 (+0.1234)	69.50 (-0.09)	61.76 (-0.72)	93.90 (+3.86)	0.4653 (+0.0168)	69.31 (-0.28)	60.98 (-0.32)	98.17 (+0.61)	0.4834 (+0.0036)
	GMR + Multi	63.51 (-2.42)	63.70 (-4.26)	53.86 (+0.40)	0.2826 (-0.0022)	70.85 (+1.26)	63.34 (+0.86)	91.67 (+1.63)	0.4564 (+0.0079)	69.88 (+0.29)	61.69 (+0.39)	96.54 (-1.02)	0.4747 (-0.0051)
	GMR + Multi + GEQ Frozen	67.28 (+1.35)	63.17 (-4.79)	74.59 (+21.13)	0.3778 (+0.0930)	69.79 (+0.20)	61.92 (-0.56)	94.51 (+4.47)	0.4666 (+0.0181)	68.44 (-1.15)	60.33 (-0.97)	97.97 (+0.41)	0.4799 (+0.0001)
	GMR + Multi + GEQ Full	69.69 (+3.76)	61.87 (-6.09)	94.31 (+40.85)	0.4653 (+0.1805)	68.92 (-0.67)	60.57 (-1.91)	98.98 (+8.94)	0.4841 (+0.0356)	68.24 (-1.35)	59.98 (-1.32)	99.59 (+2.03)	0.4862 (+0.0064)
CG-DETR	**GMR (baseline)**	**70.46**	**62.14**	**96.75**	**0.4784**	**69.98**	**61.38**	**99.19**	**0.4876**	**69.98**	**61.33**	**99.59**	**0.4895**
	GMR + GEQ Frozen	68.05 (-2.41)	63.95 (+1.81)	75.00 (-21.75)	0.3825 (-0.0959)	69.50 (-0.48)	61.99 (+0.61)	92.48 (-6.71)	0.4592 (-0.0284)	70.17 (+0.19)	61.54 (+0.21)	99.19 (-0.40)	0.4881 (-0.0014)
	GMR + GEQ Full	67.37 (-3.09)	63.18 (+1.04)	75.00 (-21.75)	0.3810 (-0.0974)	70.08 (+0.10)	61.73 (+0.35)	97.36 (-1.83)	0.4804 (-0.0072)	69.11 (-0.87)	60.62 (-0.71)	99.80 (+0.21)	0.4899 (+0.0004)
	GMR + Multi	65.25 (-5.21)	70.00 (+7.86)	46.95 (-49.80)	0.2571 (-0.2213)	68.73 (-1.25)	64.63 (+3.25)	75.41 (-23.78)	0.3845 (-0.1031)	70.17 (+0.19)	62.94 (+1.61)	90.45 (-9.14)	0.4507 (-0.0388)
	GMR + Multi + GEQ Frozen	67.57 (-2.89)	64.55 (+2.41)	70.33 (-26.42)	0.3598 (-0.1186)	69.59 (-0.39)	63.39 (+2.01)	85.16 (-14.03)	0.4273 (-0.0603)	68.92 (-1.06)	61.46 (+0.13)	92.68 (-6.91)	0.4586 (-0.0309)
	GMR + Multi + GEQ Full	66.60 (-3.86)	66.74 (+4.60)	59.15 (-37.60)	0.3115 (-0.1669)	69.11 (-0.87)	63.69 (+2.31)	81.30 (-17.89)	0.4097 (-0.0779)	68.73 (-1.25)	61.11 (-0.22)	93.90 (-5.69)	0.4637 (-0.0258)

Green: gain Red: decrease Gray: unchanged; all deltas are relative to GMR.

N-acc = TN / negatives. τ is the existence threshold. Localization uses positive ungated candidates.

Full dataset and GEQ Full training mode are distinct. Coverage and temporal limitations: see DATA.md.

Same Standard-trained checkpoints across both Test domains. Training budgets and native seeds differ; see TRAINING.md.